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# Install required library (already available in Colab)
import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Read image
image = cv2.imread('8.jpg')

# Resize for display (optional)
image = cv2.resize(image, (500, 350))

# Generate Gaussian noise
mean = 0
sigma = 25

gaussian = np.random.normal(mean, sigma, image.shape).astype(np.int16)

# Add noise to image
noisy = image.astype(np.int16) + gaussian
noisy = np.clip(noisy, 0, 255).astype(np.uint8)

# Apply Gaussian Blur
gaussian_blur = cv2.GaussianBlur(noisy, (5,5), 0)
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# Apply Median Filter
median = cv2.medianBlur(noisy, 5)

# Apply Bilateral Filter
bilateral = cv2.bilateralFilter(noisy, 9, 75, 75)

# Display images
print("Original Image")
cv2_imshow(image)

print("Noisy Image")
cv2_imshow(noisy)

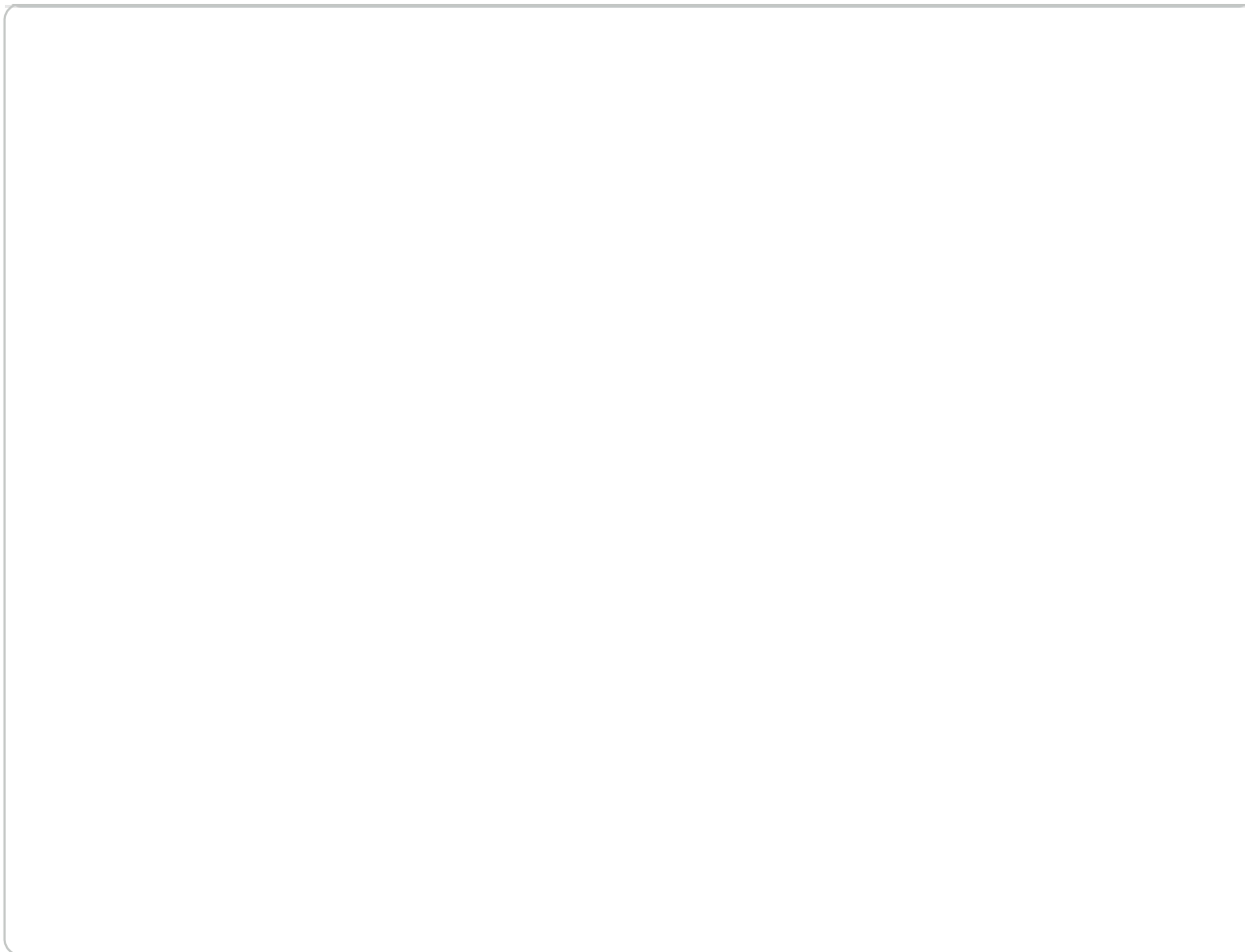
print("Gaussian Blur Result")
cv2_imshow(gaussian_blur)

print("Median Filter Result")
cv2_imshow(median)

print("Bilateral Filter Result")
cv2_imshow(bilateral)

# Save outputs
cv2.imwrite("Noisy_Image.jpg", noisy)
cv2.imwrite("Gaussian Blur.jpg", gaussian_blur)
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cv2.imwrite("Median_Filter.jpg", median)  
cv2.imwrite("Bilateral_Filter.jpg", bilateral)
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Original Image



Noisy Image





Gaussian Blur Result

